

SELECTED BODY TEMPERATURE AND THERMOREGULATORY BEHAVIOR IN THE SIT-AND-WAIT FORAGING LIZARD *PSEUDOCORDYLUS MELANOTUS MELANOTUS*

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ABSTRACT: We investigated the thermoregulatory abilities and behavior of *Pseudocordylus melanotus melanotus* (Drakensberg crag lizard) in terms of the relationship between the operative temperature (T_e), selected temperature (T_{sel}), set-point range (T_{set}) and field active body temperature (field T_{sel}), exposure to low temperature, body posture and activity. The T_e range for *P. m. melanotus* was about 58 C (–3.20 C in winter to 54.94 C in summer). In a laboratory thermal gradient, in a setting that is independent of ecological costs or thermal constraints, lizards maintained T_{set} (defined as the interquartile range of T_{sel} , after Hertz et al., 1993) between 29.00 ± 0.36 C and 31.78 ± 0.16 C in winter and 29.61 ± 0.28 C and 32.47 ± 0.18 C in summer. The mean T_{sel} was 30.08 ± 0.14 C in winter and 30.99 ± 0.11 C in summer. In the field, however, lizards achieved significantly lower T_b , which suggests that the thermal environment limited the T_b that lizards were able to achieve. Lizards were active for significantly longer and selected significantly higher T_b in summer than in winter. During winter, lizards spent a significant amount of time at T_b below their lower critical limiting temperature (defined by loss of righting). The most frequently assumed body postures in summer were those where the head or body were raised, whereas, in winter, lizards usually lay with head and body flat on the rock substrate. We suggest that these differences reflect the physiological requirements of the lizards: Head-up postures in sit-and-wait foragers are consistent with scanning for prey while head-down postures are likely motivated by thermoregulatory needs. It is clear that *P. m. melanotus* can thermoregulate efficiently, but the T_b maintained may be constrained by the range of T_e available to the lizards in their natural environment. *Pseudocordylus m. melanotus* currently appears to be geographically constrained by low environmental temperatures at the edge of its range. Should global warming become a reality in southern Africa, this species could inadvertently benefit by occupying new habitat that was previously unavailable because of thermal constraints.

Key words: Lizard; Operative temperature; *Pseudocordylus*; Selected temperature; Set-point range; Sit-and-wait forager; Thermoregulation; Thermoregulatory behavior.

CAREFUL regulation of body temperature (T_b) reduces the risk of exposure to extreme temperatures that may be lethal, and also increases the duration spent at physiologically favorable T_b (Huey et al., 1989). Physiological benefits are maximized in an ideal environment where there are few or no environmental constraints, and lizards will usually select optimal T_b when active (Huey and Slatkin, 1976). In natural settings, however, the thermal environment is heterogeneous and lizards thermoregulate by shuttling between hot and cold microclimates (or between sunlight and shade), through posture modifications (such that the surface area exposed to heat sources is altered) and by regulating

activity times (Huey, 1974). The T_b of lizards is thus dependent on both the variation in environmental temperature and on their ability to regulate heat exchange (Beck, 1996; Carrascal et al., 1992; De Witt, 1967; Peterson, 1987; Tosini et al., 1992).

Many diurnal heliothermic lizards are able to regulate T_b at high levels if they bask in sunlight during the day (Bennett, 1980). The range of T_b selected (T_{sel}) is generally considered to be the range at which the lizard can most effectively capture prey, escape predators, dig nest holes, engage in social behavior or undertake any energetically demanding activity (Bartholomew, 1977), and usually refers to the T_b adopted in laboratory thermal gradients. Hertz et al. (1993) suggest that estimates of the upper and lower set points for T_{sel} should be taken as the temperatures representing the interquartile range (mid-50% of observed selected temper-

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atures; set-point range; T_{set}). The T_b selected has also been shown to be affected by numerous factors (Huey, 1982) including reproductive condition (Andrews et al., 1997; Beuchat, 1986; Gibbons and Semlitsch, 1987; Rock et al., 2000; Rock et al., 2002; Schwarzkopf and Shine, 1991), gender (Gibbons and Semlitsch, 1987; Huey and Pianka, 2007; Rock et al., 2000; Rock et al., 2002), digestive state (Beck, 1996; Gibbons and Semlitsch, 1987), risk of predation (Downes and Shine, 1998; Shah et al., 2004), social factors (Downes and Shine, 1998), and season (Christian and Bedford, 1995; Gibbons and Semlitsch, 1987; Pentecost, 1974).

Behavioral modification of heat flux can be effected through the modification of basking frequency or duration, regulation of activity times, changes in body posture and modification of microhabitat use (Bauwens et al., 1996; Carrascal et al., 1992; Hertz, 1992; Hertz and Huey, 1981; Huey and Pianka, 1977; Huey and Slatkin, 1976; Melville and Swain, 1997; Muth, 1977; Waldschmidt, 1980; Willmer et al., 2005). Generally, when the environment is cool, lizards will select more exposed basking sites, and more shaded sites during warm periods (Cowles and Bogert, 1944; Huey and Pianka, 1977; Sherwood et al., 2005), and shuttling pattern is adapted to the use of particular basking sites, non-basking retreats, and shade (Spellerberg, 1972). A lizard will utilize the available microclimates which fall within their preferred thermal range and may only use a certain proportion of the available microclimates (Angert et al., 2002; Grant and Dunham, 1988). The duration spent thermoregulating depends on the thermal properties and availability of microclimates (Gvoždík, 2002).

Postural changes alter the body's orientation to the sun (Bartholomew, 1977; Walsberg, 1992) and body position to control the surface area exposed to solar radiation (Peterson, 1987). Although changes in body posture affect radiative heat exchange, they have the greatest effect on rates of conductive and convective heat exchange when the difference between air and substrate temperature is greatest, usually during the hottest time of day (Roberts et al., 1993). During periods of inactivity in retreats, T_b is largely

determined by air and substrate temperature, which limits thermoregulatory options (Adolph and Porter, 1993).

Diurnal lizards tend to be more active during times when environmental temperatures are optimal (Martín and Salvador, 1995). Duration of activity periods are typically longer (Adolph and Porter, 1996) and bouts of activity usually bimodal (Firth and Belan, 1998; Foà and Bertolucci, 2001; Gannon and Secoy, 1985) in summer. Generally, in winter, activity can be expected to be sporadic (e.g., *Podarcis sicula*; Foà et al., 1992) and, in spring and autumn, activity is usually unimodal (e.g., *Podarcis sicula*, Foà and Bertolucci, 2001), especially in temperate species.

Pseudocordylus melanotus melanotus (Draekensberg crag lizard) is a strictly saxicolous, diurnal, cordylid lizard. Since most rocky outcrops on which the lizard occurs are spatially heterogeneous, it is likely that they are also a highly heterogeneous environment from a thermal perspective. *Pseudocordylus m. melanotus* is an extreme sit-and-wait forager (Cooper et al., 1997), suggesting either distinct thermoregulatory behavior in terms of activity time, postures and positions, or wide thermal preferences because it spends a large portion of its day in exposed positions. Here, we (1) measured the T_{set} and T_{sel} of the lizard in the laboratory where there are few thermoregulatory constraints and minimal stress, (2) measured the field achieved T_b with respect to T_{sel} and evaluated the lizards ability to thermoregulate, (3) measured the range of operative temperatures (T_e) available to lizards in order to quantify the thermal environment, (4) quantified thermoregulatory behavior in terms of activity and body posturing for *P. m. melanotus*, and (5) assessed the risk of exposure to low temperatures in the field.

MATERIALS AND METHODS

Study Animal and Site

Pseudocordylus m. melanotus is a member of the Cordylidae, a family endemic to Africa, is saxicolous, and occur on rocky outcrops (Branch, 1998), where individuals are conspicuous when perching on rocks (McConnachie and Whiting, 2003). This species exhibits

well-developed sexual dimorphism such that males are larger and more colorful than females (Mouton and van Wyk, 1993). Snout-vent length (SVL) of adults ranges between 80 and 120 mm, but may reach a maximum of 143 mm in males (Branch, 1998).

All lizards used in this study originated from the Suikerbosrand Nature Reserve (SNR), approximately 40 km southeast of Johannesburg, South Africa (26° 27'–26° 34' S, 28° 09'–28° 21' E; 1800 m a. s. l.). The habitat in this area is typically Highveld Grassland (Rutherford and Westfall, 1986) and is dominated by *Eragrostis*, *Hyparrhenia*, *Themeda*, and *Setaria* species (Panagos, 1999). Numerous rocky outcrops suitable for *P. m. melanotus* occur in the higher altitude parts of the reserve. Rainfall is less than 500 mm per year and is summer seasonal. The climate is typically temperate; winters are cold and dry, with frequent frosts and very occasional snow, while summer temperatures frequently exceed 30 C.

The lizards were housed individually in 300 × 300 × 500 mm glass terraria with a tile shelter. Food (mealworm larvae and beetles; *Tenebrio* sp.) and water were supplied *ad libitum*. Lizards were maintained in a temperature controlled room (28 ± 1 C) with a light:dark cycle of 12:12 h.

Selected Body Temperature and Set-Point Range

Laboratory study.—Lizards were separated into three categories based on the length of time they had been in captivity. The first category of lizards had been in captivity for 12 to 18 mo for the winter and summer measures, respectively. Category two lizards had only been acclimated in captivity for between two weeks (winter) and six months (summer). The third category lizards were captured for the summer study only and were therefore only acclimated for two weeks before measures. The T_{sel} of captive lizards was measured in a thermal gradient in the laboratory in winter ($n = 32$ lizards, 2 categories) and summer ($n = 43$ lizards, three categories). Individuals of this species have previously been maintained in captivity in excess of three years (S. McConnachie, pers. obs.; lizards captured as adults) and appear to adapt well to

captive conditions. Duration spent in captivity (three categories) had no significant effect on T_{sel} (repeated measures ANOVA, $F_{12,780} = 1.11$, $P = 0.35$).

The thermal gradient consisted of a 1 × 1.5 × 0.5 m wooden enclosure. The gradient was divided lengthways into three separate compartments. Heat was provided at one end of each compartment by a 250 W infrared lamp. Cooling was achieved with a copper cooling plate connected to a water bath at the opposite end of the enclosure. A single lizard was placed in each compartment (only three lizards could be tested at a time). Since individuals could not be tested simultaneously, the experiment was repeated 11 times in winter and 15 times in summer. Lizards were placed in the thermal gradient overnight before the trials began and the lamps were turned on 30 min before the first reading was taken. Body temperature was recorded every hour (from 0900 h to 1600 h) for two days by inserting a thermocouple probe approximately 10 mm into the cloaca (after Sievert and Hutchison, 1989). The T_{set} was estimated for each lizard by determining the interquartile range of T_b selected in the thermal gradient (after Hertz et al., 1993). T_{sel} for each lizard was calculated as the mean T_b selected in the thermal gradient.

Field study.—Field work was conducted at Suikerbosrand Nature Reserve (SNR) during February–March ($n = 18$ lizards; summer) and May–June 2004 ($n = 20$ lizards; winter). All lizards were captured by noosing, individually marked using color-coded plastic collars and toe clipping, measured (SVL, nearest 1 mm; head length, head width, nearest 0.01 mm) and weighed (nearest 1 g). No lizards captured for the summer study were used in winter. In order to measure T_b in the field, small temperature data loggers (17 mm diameter, 6 mm thick; thermochron iButtons®; Dallas semiconductor, Texas, USA) were surgically implanted into each lizard. Lizards were anesthetized using 2% isoflourane gas and a small incision was made into the peritoneal cavity. The data loggers, which were pre-programmed to record body temperature every 20 min, were coated with wax and sterilized with hibicol. Data loggers weighed three grams and were less than

TABLE 1.—Summary of mean body size (mass [g] and snout-vent length [SVL, mm]) and selected body temperature (T_{sel}) for winter ($n = 32$ lizards) and summer ($n = 43$ lizards) for the lizard *Pseudocordylus m. melanotus*. Means presented $\pm$ SE.

Season		Winter	Summer
Mass	Males	45.53 $\pm$ 3.14	50.02 $\pm$ 2.11
	Females	30.62 $\pm$ 2.66	35.13 $\pm$ 1.81
	All	39.47 $\pm$ 2.50	42.76 $\pm$ 1.83
SVL	Males	117.05 $\pm$ 2.57	119.35 $\pm$ 1.77
	Females	104.38 $\pm$ 3.28	107.68 $\pm$ 2.03
	All	111.91 $\pm$ 2.28	113.67 $\pm$ 1.63
T_{sel}	Males	30.23 $\pm$ 0.17	30.68 $\pm$ 0.15
	Females	29.86 $\pm$ 0.22	31.30 $\pm$ 0.16
	All	30.08 $\pm$ 0.14	30.99 $\pm$ 0.11

10% of lizard body mass (see Table 1). The wax-coated data logger was inserted into the body cavity through the incision, which was sutured, cleaned and covered with TegadermTM. Lizards were released at the point of capture at SNR within four days of capture; all lizards had a recovery period—at least 36 h—after surgery, prior to being released.

Lizards were recaptured (also by noosing) after a minimum of three weeks and data loggers were removed using similar procedures to those used in implantation. Of the 38 lizards with implanted data loggers (18 summer and 20 winter), 22 lizards were recaptured (11 each for summer and winter), resulting in an overall recapture success of 58%. All lizards were returned to the point of capture at SNR after removal of the data loggers and identification collars. The wax coating on the data loggers was removed and data were downloaded and compared in terms of field T_{sel} and exposure to low temperatures. Operative temperatures were measured at SNR concurrently for comparison with recorded body temperatures (see below).

Field T_{sel} was calculated for each lizard over the period recorded on the implanted data loggers (approximately 28 days) and was calculated as the mean T_b over the period when the lizards were active. This period started with the highest T_b reached in the morning and ended at the highest T_b before a three-hour continuous decrease (10 consecutive decreasing T_b measures) in T_b in the afternoon or evening (Fig. 1). The three-hour decrease allowed the inclusion of shorter periods of reduced T_b during the day when

lizards may have retreated into crevices (see Alexander, 2007, for a more detailed justification of this method). Field T_{sel} was related to T_{set} by calculating the time when lizard T_b fell within T_{set} (calculated from laboratory measures).

The first three days of data were excluded to remove possible effects of surgery or stress. Days on which lizards did not emerge from their crevices (i.e., when T_b co-varied with crevice T_e) were also excluded from the analysis.

Operative Temperature

In order to select an appropriate model for measuring operative temperature (T_e), two painted (black or white), 100 mm lengths of 28 mm diameter copper pipe models (filled with water or empty) were empirically compared to the T_b of live lizards in the laboratory. We calibrated models by placing them in a 300 $\times$ 200 $\times$ 250 mm terrarium with a lizard (male, mass = 43.4 g, SVL = 120 mm). Thermocouple wires were inserted approximately 10 mm into the models and separately into the lizard's cloaca and attached to a data logger (MC Systems, 120-02 EX; Cape Town), which recorded lizard T_b and the temperature of the models every minute for one hour under various heating and cooling regimes. The lizard was monitored throughout to ensure that it was not experiencing discomfort and behaved normally. Measures of T_b were then compared to

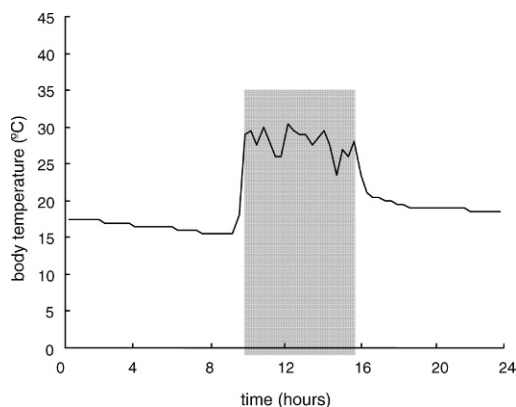


FIG. 1.—Example of field body temperature measured over 24 h using a data logger and indicating the range (grey area) over which the mean selected body temperature was calculated for *Pseudocordylus m. melanotus*.

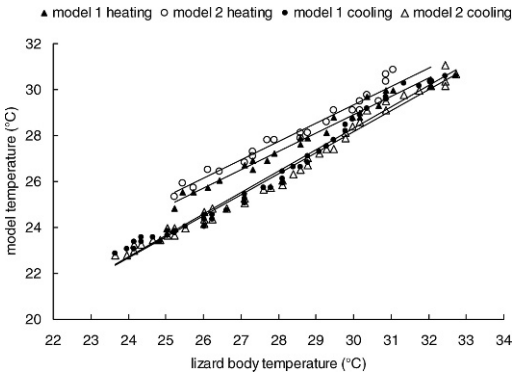


FIG. 2.—Relationship between lizard body temperature and temperature in two copper tube models with the closest representative thermal properties of *Pseudocordylus m. melanotus* (100 mm of 28 mm diameter copper pipe, painted black and filled with water). Model 1: heating, $r^2 = 0.97$, $y = 0.78x + 5.43$; cooling, $r^2 = 0.97$, $y = 0.76x + 4.62$. Model 2: heating, $r^2 = 0.88$, $y = 0.70x + 7.91$; cooling, $r^2 = 0.98$, $y = 0.75x + 4.82$. $P < 0.001$ in all cases.

measures of T_e using regression analysis. Hertz et al. (1993) suggest that, where heating and cooling rates of models and small ectotherms are similar, comparisons of T_b and T_e will give an indication of whether the animals are actively thermoregulating. Therefore, the lizard models were considered good T_e models based on the high R^2 values and the slope of the regression was close to 1. The model that proved to be the best indicator of T_e representative thermal properties for *P. m. melanotus* was the black-painted copper pipe filled with water ($r^2 = 0.97$ for heating and cooling; slope = 0.78 and 0.76 for heating and cooling, respectively; Fig. 2).

Lizard T_e was measured in the field at SNR between summer 2002 and winter 2004. Three models (black, 100 mm lengths of 28 mm diameter copper pipe filled with water) were placed at a single locality in the field and connected to a data logger (MC Systems, 12-02 EX; Cape Town), which was programmed to log temperatures of the models every 20 min. Models were placed individually in microhabitats where lizards would be expected to equilibrate at the lowest and highest T_b : In a crevice (approximately 100 mm deep with 150–200 mm of rock on either side; limited by the size and shape of the model) and on the rock surface

such that one model received morning sunshine and the other afternoon sunshine. Records from the models were used as measures of the highest and lowest temperatures available to the lizards. The T_e was related to T_{set} by calculating the time T_e fell within T_{set} (calculated from laboratory measures).

Thermoregulatory Behavior

Behavioral thermoregulation in *P. m. melanotus* was quantified in terms of the duration that lizards were active and the frequency that each body posture was assumed by lizards during a specific period. Activity time was quantified by inference: T_b profiles were measured using implanted thermochron iButtons and duration of activity was taken as the time from the first rapid increase in T_b until the time of the highest temperature before a three-hour continuous decrease in T_b . Initiation of activity in the morning was easily diagnosed by a rapid increase in T_b when the lizard first emerged and began basking. Once the T_b reached the plateau phase, T_b was kept in a narrow range by the lizard.

Body posture and position were recorded during focal animal sampling in the field at SNR. Individual lizards were observed continuously for 30 min in summer when they were encountered in the field. Since lower levels of activity were expected in winter, observations were 60 min to ensure sufficient observation time during winter. Owing to their strict site fidelity and distinct color pattern, some individuals were observed on more than one occasion. Observation times for lizards observed on more than one occasion were summed. Body postures and positions assumed by the lizards were categorized as follows (Fig. 3):

- Prostrate (head and body flat on the rock surface).
- Prostrate with head raised.
- Upper body raised (front of body and head raised, forelegs partially or fully extended, abdomen and tail flat on rock surface).
- Body raised (whole body raised off rock surface, all legs partially or fully extended).

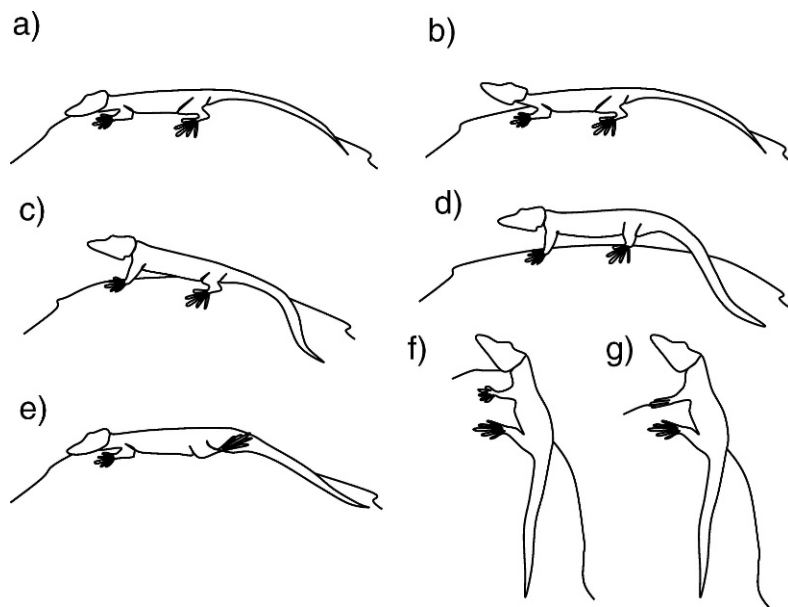


FIG. 3.—Categories of body postures and positions assumed by *Pseudocordylus m. melanotus*. (a) prostrate; (b) prostrate with head raised; (c) upper body raised; (d) body raised; (e) legs raised; (f) side of rock; (g) on side of rock extended. See text for details.

- (e) Legs raised (body and head flat, hind legs and feet raised off rock surface).
- (f) Side of rock (on side of rock, head extended above top of rock).
- (g) Side of rock extended (on side of rock with head and upper body extended above top of rock, forelegs partially or fully extended, sometimes body raised off rock surface).
- (h) Other (any other body posture or position assumed).

In summer, 122 focals were recorded for 92 individual lizards (53 male and 39 female). In winter, 117 focals were recorded for 50 individual lizards (28 male and 22 female). Not all lizards were observed for the entire 30 or 60 min focal duration since lizards were occasionally obscured from view by rocks or vegetation and one to four focals were conducted per lizard. Mean ($\pm$ SE) observation time per lizard in summer was 34.76 ± 2.59 min and ranged between two and 120 min and the total, overall observation time in summer was 51 h 55 min. Mean ($\pm$ SE) observation time per lizard in winter was 110.02 ± 25.76 min and ranged between two and 1066 min and the total, overall

observation time in winter was 91 h 41 min. Focal sampling started at 0700 h (GMT + 2) and continued through the day until no more lizards were observed; this was usually between 1700 h and 1800 h (Fig. 4). In winter, the first lizards were observed between 0900 h and 1000 h.

Exposure to Low Temperatures

Exposure to low temperatures was calculated as the number of days in an average 30-day period that the lizards experienced periods where their T_b was at or below 10 C (below lower critical limiting temperature, CTMin; see McConnachie et al., 2007). Lizards were incapacitated at any temperatures below CTMin and had compromised locomotory ability (McConnachie et al., 2007). The proportion of time at or below 10 C was also determined as a measure of the time spent at or below 10 C. During summer, no lizards experienced periods where their T_b was below 10 C.

Statistical Analyses

Data were analyzed using Excel and STATISTICA 5.5[®] and all tests were conducted at a 5% level of significance. All data were tested

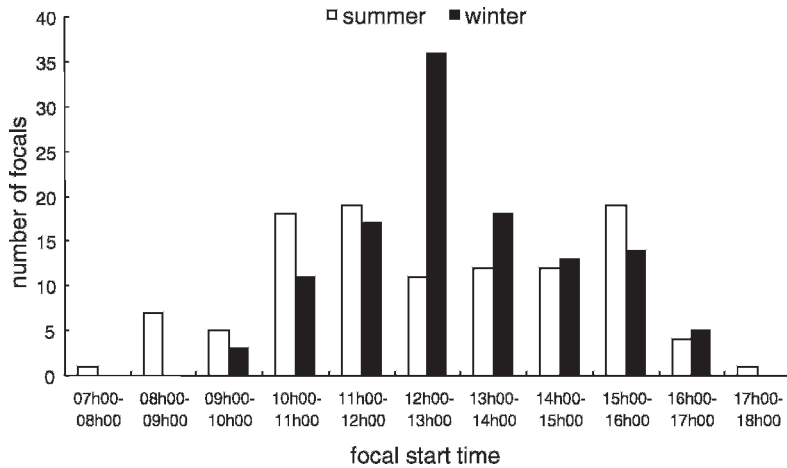


FIG. 4.—Temporal spread of focal starting times from when the first *Pseudocordylus m. melanotus* was observed in the morning (0700 h) until the last lizard was observed in the afternoon (between 1700 h and 1800 h); summer = open, winter = black.

for normality (Lilliefors' test) before applying parametric statistics. All means are presented $\pm$ standard error (SE).

RESULTS

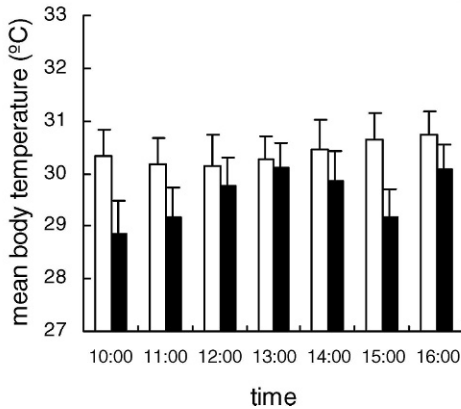
Selected Body Temperature and Set-Point Range

Laboratory study.—Mean T_{set} was 29.00 ± 0.36 to 31.78 ± 0.16 C and 29.61 ± 0.28 to 32.47 ± 0.18 C in winter and summer, respectively. Body sizes and mean T_{sel} of males and females are summarized in Table 1. The mean T_{sel} for all lizards was 30.08 ± 0.14 C and 30.99 ± 0.11 C for winter and summer, respectively. Male and female lizards did not select significantly different temperatures between seasons (repeated measures ANOVA with gender and season as factors: $F_{7,903} = 0.51$, $P > 0.5$). Post hoc, lizards selected significantly higher temperatures in summer ($t_{71} = 4.27$, $P < 0.001$). In both summer and winter, body size and T_{sel} were not significantly related (SVL and body mass, $P > 0.05$ and $r^2 < 0.1$). Overall, difference in T_{sel} between time intervals were not significantly different (ANOVA, winter, $F_{6,190} = 0.20$, $P = 0.97$; summer, $F_{6,195} = 0.85$, $P = 0.53$). No pattern of T_{sel} was evident during photophase for either winter or summer measures (ANOVA, winter, $F_{1,6} = 0.20$, $P = .97$; summer, $F_{1,6} = 0.86$, $P = 0.53$; Fig. 5).

Field study.—Body mass and SVL were significantly correlated ($r = 0.92$, $P < 0.001$). Snout-vent length of male and female lizards in the summer and winter groups were not significantly different (ANOVA, $F_{3,18} = 2.55$, $P = 0.09$). In summer, larger individuals attained significantly lower field T_{sel} (regression analysis, $P = 0.008$, $r^2 = 0.58$; Fig. 6). Field T_{sel} and body size in winter were not significantly related (regression analysis, mass, $P = 0.63$, $r^2 = 0.03$; SVL, $P = 0.77$, $r^2 = 0.01$).

On days where lizards were actively thermoregulating, lizard T_b followed a very different pattern to the minimum and maximum T_e (Fig. 7). Mean ($\pm$ SE) field T_{sel} during summer was 28.91 ± 0.26 C, and during winter was 26.30 ± 0.47 C. Lizards attained significantly higher temperatures in summer than winter (ANCOVA with gender, mass and SVL as covariates, $F_{1,17} = 20.88$, $P < 0.001$). Time where T_b was within T_{set} did not differ significantly between male and female lizards during summer or winter (summer, $t_{268} = 1.97$, $P = 0.68$, $n = 5$ females, 6 males; winter, $t_{198} = 1.97$, $P = 0.75$, $n = 4$ females, 7 males; Fig. 8). Lizards spent significantly longer periods in summer where their T_b was within T_{set} than during winter ($t_{20} = 2.09$, $P < 0.001$, $n = 22$ lizards; summer, 179.00 ± 13.65 min/day; winter, 55.50 ± 7.89 min/day).

(i) Winter



(ii) Summer

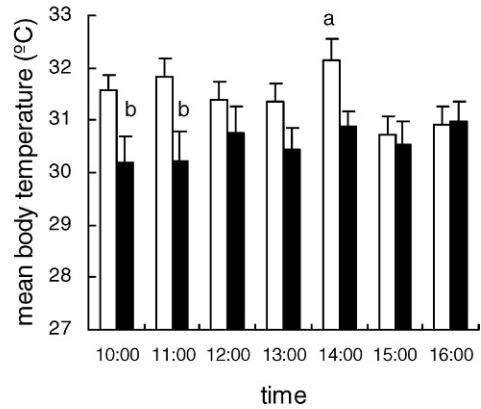


FIG. 5.—Mean selected body temperature ($\pm$ SE) of lizards in a thermal gradient between 1000 h and 1600 h during (a) winter ($n = 32$ lizards) and (b) summer ($n = 43$ lizards). Letters indicate significant differences between days and time intervals (Post hoc Tukey $P < 0.05$).

Operative Temperature

During winter, crevice temperature rarely exceeded 10 C, and the lowest recorded T_e in a crevice was -4.25 C, recorded during July 2002. For 2004, during the period when lizards carried implanted thermochron iButtons and were active in the field, operative temperatures varied between -3.20 C and 54.94 C (in crevice and outside in the sunshine; Table 2). Periods where T_e fell within T_{set} were significantly longer in winter than in summer ($t_{56} = 2.00$, $P = 0.04$, $n = 58$ days; summer, 46.90 ± 6.22 min/day; winter, 74.48 ± 12.27 min/day).

Thermoregulatory Behavior

Lizards were active for significantly longer during summer than in winter ($t_{20} = 9.96$, $P < 0.001$). In summer, lizards emerged at 0816 h $\pm$ 0011 h (GMT + 2) and were active for 8 h 51.84 min $\pm$ 20.80 min, returning to their retreats at 1707 h $\pm$ 0010 h. Summer activity duration did not differ significantly between males and females ($t_9 = 0.05$, $P = 0.96$). In winter, lizards emerged at 1048 h $\pm$ 0041 h and were active for 4 h 20.61 min $\pm$ 17.59 min, returning to their retreats at 1431 h $\pm$ 0013 h, and did not differ between males and females ($t_9 = 0.43$, $P = 0.68$). Lizards were active for significantly more days in summer than in winter ($t_{20} = -4.05$, $P = 0.001$), and were active every day in summer

and on 21.71 ± 2.05 days out of an average 30-day period in winter (i.e., $72.36\% \pm 0.68$ of winter days).

Lizards shuttled between rock crevices, full sunshine and areas in the shadows of rocks, or vegetation adjacent to the rock. Lizards tended to either face the sun directly, face directly away from the sun thus receiving full sunshine on their backs, or turned side-on to the sun. Although males and females exhibited similar postures and microhabitat selection, there were clear seasonal differences. The most frequently assumed postures in summer were (b) prostrate with head raised and (c)

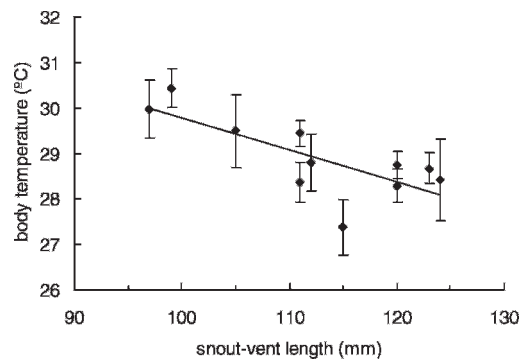
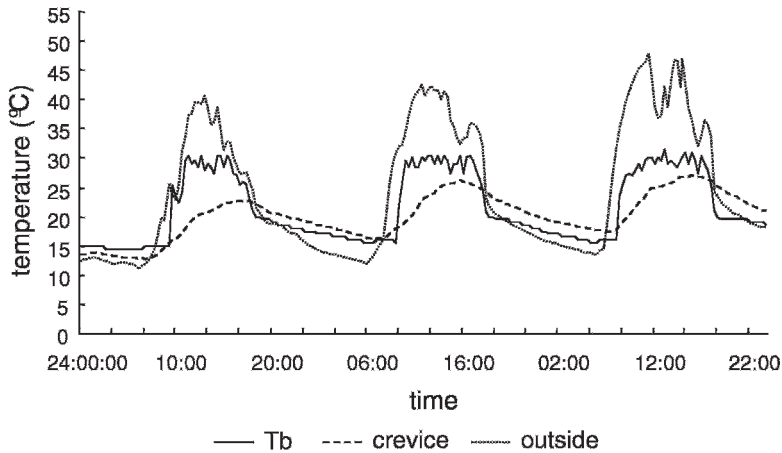


FIG. 6.—The relationship between selected body temperature (T_{sel}) in the field and snout-vent length during summer for *Pseudocordylus m. melanotus* ($r^2 = 0.58$, $P = 0.008$, $y = -0.07x + 36.79$).

a) summer



b) winter

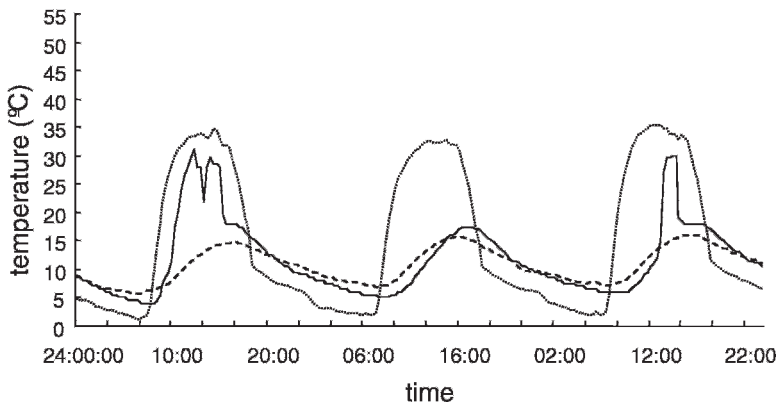


FIG. 7.—Example of field active body temperature (T_b) of a lizard (solid line) relative to operative temperatures (T_e ; dotted lines) over an average three-day period (each peak indicates a day) during (a) summer and (b) winter at Suikerbosrand Nature Reserve. Active thermoregulation is evident in the difference between the measured T_b and T_e inside a crevice and outside in the sunshine. During the night lizard T_b is buffered by rock and therefore does not drop to the T_e outside its retreat. During the day (peaks), lizards actively thermoregulate such that T_b does not match the T_e . In winter (b), lizards are active for short periods only (indicated by narrower peaks in T_b) and, in this case, the lizard does not leave its retreat (indicated by the peak in T_b mirroring that of the peak in crevice temperature).

upper body raised, and (d) legs raised and other postures and positions were only observed during focal animal sampling in the afternoon (Fig. 9). In winter, the most frequently assumed postures were (a) prostrate, (c) upper body raised and (f) side of rock. Other postures observed were only observed in the afternoon, including (b) head raised and (g) side of rock extended (Fig. 9).

Exposure to Low Temperatures

In an average 30-day period in winter, lizards experienced 27.91 ± 1.24 days with periods of $11 \text{ h } 1.8 \text{ min} \pm 1 \text{ h } 46.2 \text{ min}$ (33.51 ± 5.41 20 min intervals) spent at or below 10°C (CTMin). This means that lizards are spending approximately 47% of their time at or below their CTMin during winter, and were therefore incapacitated for a significant

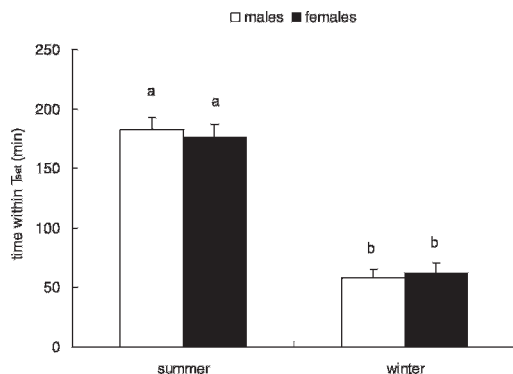


FIG. 8.—Time where field body temperature fell within the set-point range (T_{set}) measured in the laboratory during summer and winter for male and female *Pseudocordylus m. melanotus*. Letters indicate significant difference between seasons, but not between genders within seasons ($P < 0.05$).

portion of the day. The lowest T_b experienced by lizards during summer was 14.15 ± 0.15 C, while the lowest T_b experienced by lizards during winter was 7.27 ± 0.38 C.

DISCUSSION

The operative temperature range for *P. m. melanotus* was -3.20 C to 54.94 C in the coldest and hottest microclimates, respectively. This means that lizards had a wide range of environmental thermal opportunities over the seasons (although not necessarily at one instant in time). The T_b measured in the laboratory thermal gradient was higher than field T_{sel} , which suggests that there may be limitations to the lizards' thermal environment, particularly in winter.

During summer, lizards did not experience T_b below 10 C. The lowest T_b experienced during summer was approximately 14 C. During winter, lizards experienced periods where their T_b was below 10 C approximately 28 days out of 30, and the lowest T_b experienced was approximately 7 C. This suggests that, during summer, lizards will rarely, if ever, experience T_b below their CTMin, whereas, in winter, lizards spend a significant amount of time (47%) unable to move.

Lizards were active for significantly shorter periods during winter than during summer and activity time in winter was around half that of activity time during summer. Despite

TABLE 2.—Minimum (nighttime) and maximum (daytime) operative temperatures (C) measured inside and outside a crevice during summer and winter 2004 at Suikerbosrand Nature Reserve.

		Summer	Winter
Crevice	Min	21.8	-0.7
	Max	29.4	18.3
Outside	Min	10.6	-3.2
	Max	54.9	42.3

being active for longer periods in summer, available T_e within T_{set} was longer in winter and was also within T_{set} for a greater proportion of the activity time (33% in winter as opposed to 11% in summer). In summer, however, T_e often exceeded the upper limit of T_{set} (up to ca. 55 C). This suggests that there was greater thermoregulatory opportunity for the lizards in summer. Changes in position and posture, and shuttling between sunshine and shade were obvious thermoregulatory behaviors. These behaviors are consistent with those associated with both heliotherms and thigmotherms, thus suggesting a mixed thermoregulatory strategy in *P. m. melanotus*.

Selected T_b in many lizards generally approach 30 C (e.g., Andrews and Kenney, 1990; Angilletta et al., 1999; Arad et al., 1989; Grbac and Bauwens, 2001; Rocha and Vrcibradic, 1996). The T_{sel} measured for *P. m. melanotus* is thus within the range expected for lizards, as well as for other cordylid lizards. Selected T_b measured in thermal gradients for *Cordylus vittifer* and *C. jonesi* was 32.1 C (Skinner, 1991) and 33.5 C (Wheeler, 1986), respectively. In summer, the field T_b of *P. m. melanotus* is similar to those of other cordylids. Bauwens et al. (1999) reported the range of field T_b for *Cordylus cataphractus*, *C. macropholis*, *C. niger*, *C. polyzonus*, and *P. capensis* to be between 29 and 32 C.

The T_{sel} of male and female *P. m. melanotus* was not significantly different for either summer or winter measures. This result is not surprising: A recent review of gender differences in aspects of thermal biology among 56 species from seven lizard clades found minor differences (<1 C in mean body temperature; Huey and Pianka, 2007; also see Lailvaux, 2007 for a review). Gender difference in T_{sel} is usually explained by reproduc-

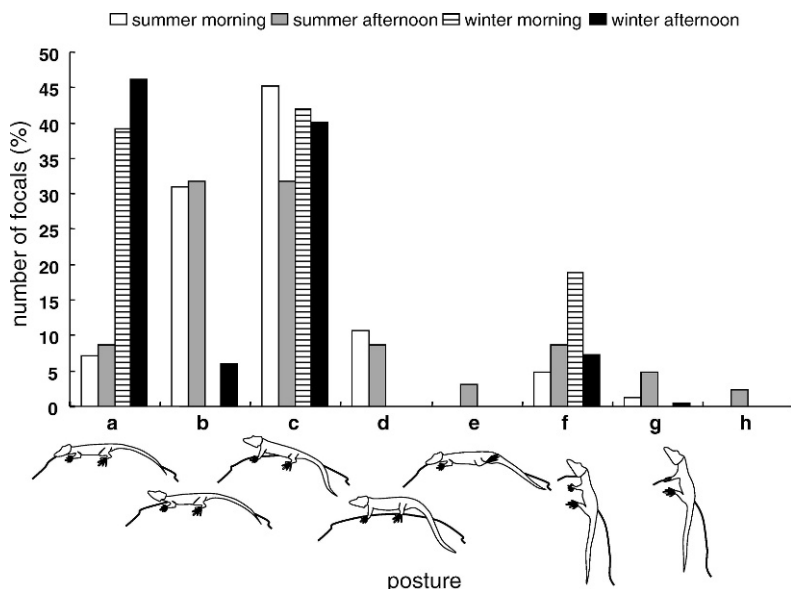


FIG. 9.—Frequency of postures and positions assumed by *Pseudocordylus m. melanotus* during the morning and afternoon in summer and winter: (a) prostrate, (b) prostrate with head raised, (c) upper body raised, (d) body raised, (e) legs raised, (f) side of rock, (g) side of rock extended, (h) other. Frequency calculated as the percentage of focals in which the posture or position was observed.

tive condition where gravid females either select higher or lower T_b than non-gravid females or males. For example, gravid female *Sceloporus jarrovi* (Beuchat, 1986) and *S. grammicus* (Andrews et al., 1997) have lower T_{sel} than non-gravid females and males, whereas, gravid female *Haplodactylus maculatus* selects higher T_b than post-parturient/non-gravid females and males (Rock et al., 2000; Rock et al., 2002). Young of *P. m. melanotus* are born during late summer (Flemming, 1993). Some females observed during the field study may therefore have been gravid. The direct effects of reproductive status on thermoregulation in this species, however, remain to be investigated.

Pseudocordylus m. melanotus selected higher T_b in summer than in winter. Christian and Bedford (1995) suggest that seasonal shifts in preferred T_b could be due to acclimatization in response to environmental temperatures, photoperiod, reduced food availability, and hormonal cycles. This has been noted in *Lacerta viridis*, where seasonal changes in T_b are not solely dependent on the availability of thermal resources, but also photoperiod, especially where food availability and thermal

regimes remain constant (Rismiller and Heldmaier, 1982, 1988). In *P. m. melanotus*, the thermal environment may not be limiting because most lizards are active during winter, even if only for a few days in a given month. Food availability could be a limiting factor since some reptiles are known to select higher T_b after eating (e.g., Beck, 1996; Brown and Griffin, 2005; Gibson et al., 1989).

The lowest temperature in the coldest microclimate (i.e., in a crevice) was -4°C in winter 2002. This is 1.15°C above the measured lower lethal temperature (-5.15°C ; McConnachie et al., 2007), and 14°C below the measured lower critical limiting temperature (CTMin ca. 10°C ; McConnachie et al., 2007). In addition, crevice temperature rarely exceeds 10°C during winter and T_b of lizards measured in the field is below CTMin for approximately 47% of the time. This suggests that lizards are significantly affected by the thermal environment in winter and are totally incapacitated, or effectively comatose, during most winter nights. Although a colder acclimation temperature may have resulted in lower CTMin values, free-ranging lizards, although they experience lower nighttime

temperatures, are selecting high T_b when they are active during winter.

The lowest recorded temperature in the coldest microclimate during summer was 1.09 C (0020 h, October 2004), suggesting that lizards do occasionally experience environmental temperatures below CTMin during summer, but for relatively short periods. The hottest recorded temperature was 54.94 C outside in full sunshine. Consequently, lizards would be able to attain significantly higher T_b than they are actually selecting. The discrepancy between the laboratory and field measures of T_{sel} may therefore be explained by variations in individuals and territory structure; for example, some lizards may hold territories which have fewer thermal resources, or restricted available microclimates, thus restricting the T_e available in any particular area.

Since measures of T_{sel} in thermal gradients are indicative of T_b selected and maintained in an environment with few or no thermoregulatory constraints, this suggests that there may be environmental constraints on the thermoregulatory ability of *P. m. melanotus*. The lizards were, however, actively thermoregulating in the field. This is evident in the discrepancy between the T_b and T_e measures (see Fig. 7). Also, the highest T_e recorded (54.94 C) is obviously too high for the lizards to withstand (see also McConnachie and Alexander, 2004), so lizards would have to avoid places where their T_b might exceed their critical limiting maximum. Body temperatures are thus maintained at a level which is determined primarily by the environment and the lizards ability to behaviorally regulate its T_b through activity, shuttling or postural changes.

Generally, activity in reptiles during cool periods is more sporadic, shorter and unimodal, while, in summer, activity is regular, longer and bimodal. For example, the prairie rattlesnake, *Crotalus viridis viridis*, shows unimodal activity in spring and autumn with a peak during midday, and bimodal activity in summer with a peak in the morning and again in the afternoon (Gannon and Secoy, 1985); the scincid lizard *Tiliqua rugosa* exhibits bimodal activity during summer also with peaks in the morning and afternoon (Firth

and Belan, 1998). On days when *P. m. melanotus* are active in winter, T_b measured in the field suggests that activity is unimodal. This is likely because lizards maintain high T_b for short periods only once during the day. In summer, high T_b is maintained throughout the day, which also suggests a unimodal activity pattern. During focal animal sampling, however, the number of lizards active during the heat of midday in summer was lower, which suggests a degree of bimodality of activity.

Even when thermal conditions are suitable, not all lizards within a population are simultaneously active (Martín and Salvador, 1995), but basking lizards are exposed, thus increasing the probability that they will be observed (Foà et al., 1992). So, although *P. m. melanotus* appear to be active, they may, in fact, seek out suitable retreat sites during the hottest times of the day. Based on differences between T_b and T_e it is clear that *P. m. melanotus* are thermoregulating throughout the day. Although they may not be active, they are selecting retreats where they can still maintain T_b at a selected level. Huey et al. (1989) suggest that retreats offer equivalent, or sometimes superior, thermoregulatory opportunities to those available in more exposed environments. These periods of inactivity may be important in avoiding environmental extremes (Kearney, 2002), conserving energy or water (Martín and López, 2000), and avoiding predators (Webb and Whiting, 2005).

Four strategies of heat regulation in lizards have been identified. These include (1) gaining external heat or avoiding heat loss to the environment, (2) retaining internal heat, (3) generating internal heat, and (4) losing excess heat or avoiding heat gain from hot environments (Muth, 1977; Sherwood et al., 2005). Generating and retaining internal heat (2 and 3) generally constitute physiological measures of heat regulation, while losing and gaining external heat (1 and 4) include behavioral thermoregulation. Sherwood et al.'s (2005) descriptions of external heat loss and heat gain are clearly evident in the behavior of *P. m. melanotus*. In gaining external heat, lizards bask in sunlight on cold mornings and gain heat from radiation and conduction from warmed surfaces. Once an

optimal level has been reached, the lizards become active, seeking food, while using behavioral and physiological methods of T_b maintenance.

The most frequently assumed postures, throughout the day, were (b) prostrate with head raised and (c) upper body raised (see Figs. 2 and 9). This suggests that lizards are effectively using both solar radiation and conduction from the rock surface (i.e., a combination of heliothermic and thigmothermic thermoregulation). Generally, lizards assumed more and/or different postures during the afternoon. During the hottest periods, if lizards were exposed, they assumed more elevated postures, with either more of the body or legs raised off the rock surface. This means that the lizards are potentially avoiding direct contact with the rock surface and thus minimizing heat gained from it, as well as increasing potential heat loss by exposing a greater surface area to the air, which may be especially important when it is windy. Also, lizards also assumed more prostrate postures during winter and when it was overcast (S. McConnachie, personal observation) thus maximizing heat gain from the rock during cool periods.

Pseudocordylus m. melanotus can clearly regulate and maintain its T_b at a selected level and body posture is undoubtedly important in thermoregulation. In *P. m. melanotus* the differences in body postures (daily and seasonally) may also reflect physiological state with head-up postures indicative of scanning for prey and head-down postures likely motivated primarily by thermoregulation needs. In summer, T_{sel} may be constrained by environmental factors because lizards did not maintain optimal T_{sel} (as measured in a thermal gradient), although they could have achieved T_b greater than T_{sel} . In winter, it is likely that there are limitations to the thermal environment, which do not allow lizards to achieve and maintain T_{sel} . In particular, a significant portion of their time (43%) is spent at temperatures below CT_{Min} , sometimes approaching to within 1.5°C of critically lethal temperatures. It therefore appears that *P. m. melanotus* is at the lower end of its thermal range and any climate warming may alleviate thermal constraints and allow for range expansion where suitable rock exists.

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